

Parth Natekar

DEPARTMENT OF ENGINEERING DESIGN, IIT MADRAS

C-1101, Beverly Hills
Pancard Club Road, Baner
Pune, Maharashtra, India 411045
Github | parthnatekar26@gmail.com
Website | [LinkedIn](#) | +91-99-609-50978

EDUCATION	INSTITUTION & DEGREE	CGPA	YEAR
	Indian Institute of Technology Madras <i>Bachelor of Technology in Engineering Design, Master of Technology in Biomedical Design</i>	8.89/10.00	2020
	Fergusson College <i>Higher Secondary School (Major: Electronics)</i>	92.4 %	2015
	Vikhe Patil Memorial School <i>Secondary School (CBSE)</i>	97.0 %	2013
PUBLICATIONS	Parth Natekar, Manik Sharma. "Representation Based Complexity Measured for Predicting Generalization in Deep Learning." <i>Winning Solution of the NeurIPS 2020 competition on Predicting Generalization in Deep Learning</i>. (Preprint). (Updated version written with runner-up teams will be published as part of the NeurIPS 2020 post-proceedings, under completion). Parth Natekar, Avinash Kori, and Ganapathy Krishnamurthi. "Demystifying Brain Tumour Segmentation Networks: Interpretability and Uncertainty Analysis." <i>Frontiers in Computational Neuroscience 14 (2020): 6</i> (Link). Avinash Kori, Parth Natekar, Balaji Srinivasan, and Ganapathy Krishnamurthi. "Abstracting Deep Neural Networks into Concept Graphs for Concept Level Interpretability." <i>Accepted to AAAI 2021 Health Intelligence Workshop, arXiv:2008.06457 (2020)</i>. (Link).		
RESEARCH & PROFESSIONAL EXPERIENCE	Anheuser-Busch InBev, Bangalore Data Scientist Machine Learning for Compliance Analytics <i>Aug '20 - Present</i> - Working on building machine learning models and developing hypotheses to flag anomalous and risky transactions for legal and financial compliance. Medical Imaging and Reconstruction Lab, IIT Madras Research Associate Interpretable Deep Learning for Medical Image Analysis (Guide: Dr. Ganapathy Krishnamurthi, Engineering Design, IIT Madras) <i>June '19 - June '20</i> - Conceptualized a pipeline for interpretability and uncertainty estimation of deep learning models which perform various image processing tasks in the medical domain, such as tumor segmentation. - This involved developing a new method for concept-level interpretability as well as implementing techniques like Network Dissection and Bayesian Dropout on segmentation and classification models. - Implemented the pipeline on benchmark brain tumor segmentation datasets to understand information flow and representation in internal layers and to quantify uncertainty of model predictions. - Developed a python package for direct implementation of the proposed interpretability pipeline on deep neural networks for any relevant biomedical dataset. Honeywell Technology Solutions, Bangalore Research Intern Graphical Modelling of Neural Networks for Causal Inference (Mentor: Satyanarayan Kar, Honeywell Aerospace, India) <i>January '19 - June '19</i> - Devised a probabilistic graphical model based algorithm for Causal Inference on a deep neural network, to provide human-understandable explanations of what the network actually learns. - Implemented the algorithm on benchmark datasets and successfully delineated variables in the model which were causal for a particular prediction compared to those with spurious correlations. - Developed a deep learning pipeline for time-series analysis of flight data in order to predict anomalous flight landings. Achieved 86% accuracy in predicting instability 2 minutes prior to landing. - Implemented an algorithm for object detection and text extraction from aircraft documents.		

Convergence Design Lab, Purdue University, USA

Research Intern | Wearables for Human-Computer Interaction

(Guide: *Dr. Karthik Ramani*, Mechanical Engineering, Purdue University) May '18-July '18

- Completed the entire design cycle – ideation, detail design, fabrication and prototyping, of IOT based flexible, wearable electronic devices for interaction with Mixed Reality environments.
- Devised a novel method to design flexible electronic circuits on various fabric substrates. Conducted material and sensor tests to integrate tactile sensing using carbon based piezoresistive sensors.
- Developed multiple ergonomic, working prototypes. Set up networking capabilities via UDP/TCP protocols and tested the wearables in Mixed Reality and IoT environments.
- Created applications in Unity and C# to validate the efficiency of using the prototyped wearables.
- Reduced task time by 44% and error rate by 74% vis-à-vis traditional interaction methods.

CERN (European Organization for Nuclear Research), Geneva

Engineering Intern | Testing Apparatus for CMS cooling system

(Mentor: *Antti Onnela*, *EP-DT*, CERN, Geneva) May '17 - July '17

- Selected to be part of a 3-member team from IIT Madras to work on the Large Hadron Collider.
- Designed an adiabatic testing apparatus for the evaporative CO₂ based cooling system of the CMS Detector using concepts from thermal, mechanical and electrical design.
- Set up sensing and feedback system for monitoring of performance of the cooling system
- Created a software interface in LabView to enable easy user interaction.

AWARDS & ACHIEVEMENTS

- **Winner, NeurIPS competition on Predicting Generalization in Deep Learning**, 2020
- **PURE fellow**: Awarded PURE Scholarship for a two-month Research Internship at Purdue University, Indiana, having been selected in the top 14 among 800 students from IIT Madras.
- **CBSE Class X School Topper**: Came first in school with percentage of 97.00%. Awards for highest marks in Math (100/100) and Science (99/100).
- One of 15 students selected from 5000 by Infosys for their ‘Spark – Catch Them Young’ Program.

RELATED COURSE WORK

Computation & Machine Learning

- Machine Learning for Engineering Applications
- Probabilistic Graphical Modelling
- Computational Systems Biology
- Computational Methods in Design
- Linear Algebra for Engineers
- Introduction to Computation and Visualization
- Data Structures and Algorithms in Python

Biomedical Design

- Medical Image Analysis
- Design of Diagnostic and Monitoring Systems
- Design of Implantable and Surgical Devices
- Medical Equipment Dissection Lab
- Human Anatomy, Physiology & Biomechanics
- Biomaterials Engineering

Mechanical/Electrical Engineering

- Design of Mechanical Systems (I & II)
- Design of Electronic Systems (I & II)
- Digital Signal Processing
- Engineering Mechanics
- Design of Thermal and Fluid Systems
- Electromagnetic Compatibility for Design
- Bio-Micro-Electrical-Mechanical Systems

Design Methodology & Art

- Functional and Conceptual Design
- Human Factors in Design
- Geometric Modelling and CAD
- Creative Design
- Design for Form and Aesthetics
- Computational Methods in Design

SKILLS

Programming: Python, C, C++, C#, JavaScript, Unity, Machine Learning (Keras, Tensorflow, Pytorch), Arduino, Raspberry Pi, L^AT_EX, SQL
Languages: Competent French, Limited Working German.

RELEVANT
COURSE
PROJECTS

Computational Modelling of Pathogen Interactomes

January to May 2018 | Supervisor : [Dr. Karthik Raman](#)

- Delineated drug-resistance pathways in the pathogen Staph Aureus through systems level computational modelling of the protein-protein interaction network of the organism.
- Devised a modified network-based shortest path algorithm by integrating node centrality measures with chemical protein interaction scores between drug molecules and proteins.
- Achieved 68.5% prediction accuracy on already known resistance pathways, validating the viability of our algorithm for exploring drug resistance pathways in pathogens.

Colour Sensing Robotic Manipulator

January to May 2017 | Supervisor : [Dr. N. J. Vasa](#)

- Designed and implemented a 6-dof color detecting robotic manipulator with a magnetic tooltip arm.
- Involved electronic circuit design with the use of a microcontroller to control the robot.
- Coded in C/C++ using the Arduino IDE.

Ultrasound Nerve Image Segmentation

Jan to May 2017 | Supervisor : [Dr. Ganapathy Krishnamurthi](#)

- A deep learning-based project for semantic segmentation of ultrasound nerve cell images.
- Used a fully-convolutional neural-network implemented in Python using Keras and Tensorflow.
- Obtained 71.2% Dice Score on test dataset – on par with then state-of-the-art techniques

TEACHING
AND
VOLUNTARY
EXPERIENCE

Teaching Assistant - Design of Electronic Systems Lab, IIT Madras

Professor : [Dr. Tuhin Subhra Santra](#)

Aug '19 - Dec '19

- Selected as Graduate Teaching Assistant for the Design of Electronic Systems Lab (Fall 2019).
- Managed and guided a group of 60+ students in electronic circuit design principles.

Teaching Assistant - Medical Image Analysis, IIT Madras

Professor : [Dr. Ganapathy Krishnamurthi](#)

Aug '19 - Dec '19

- Conducted tutorial sessions to teach students the programming aspects of Medical Image Analysis.

Deputy Head, International Relations Cell, IIT Madras

July '17 - July '18

- Led and organized a team of 20+ people, managed a group of 80+ international exchange students.
- Helped organize various events, including 'International Day', which catered to 2000+ people.

Volunteering

- Worked with an organization called *Lata Mahajan Foundation* on their project *EducationLinked*, which is aimed at overall development of tribal and underprivileged children.
- Worked with *iVolunteer*, Bangalore to help organize health camps in orphanages.

EXTRA
CURRICULAR
ACTIVITIES

Institute Football Team

Aug '16 - Dec '17

- One of only two freshers to be a part of the Institute Football Team.
- Captain of School Football Team. Participated in CBSE West Zone Tournament in 2011 and 2012

Literary and Debate

- Secured the 3rd position for Ganga Hostel, IIT Madras in the Lit-Soc inter-hostel debate competition.
- Part of organizing committee of IIT Madras Model United Nations 2017.

Art and Design

- Selected Impressionist paintings uploaded on [Behance](#).
-